

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

View Only

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QUESTIONS

Q1

ATM Withdrawal – Exception
Handling Debugging
10 marks

Q2

Hospital Registration –
NullPointerException
Debugging
10 marks

Q3

E-Commerce Product Search –
Array Exception Debugging
10 marks

Q4

Bank Account – Race Condition
Debugging
10 marks

Q5

Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q2

Hospital Registration –
NullPointerException Debugging10
marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter patient name: ");
        String inputName = sc.nextLine();

        String name;
```

```

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
}

```

The corrected program must prevent
NullPointerException and handle invalid age input
without terminating unexpectedly.

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 import java.util.*;
2 class PatientRegistration{
3     public static void main(String[] args)
4         Scanner sc = new Scanner(System.in);
5         try{
6             String inputName=sc.nextLine();
7             String name=inputName.equals('
8             int age=sc.nextInt();
9             if(name==null)
10                 System.out.println("Patier
11             else
12                 System.out.println("Patier
13                 System.out.println(age>=18?"Ac
14         }

```

```
15         catch (InputMismatchException e) {  
16             System.out.println("Invalid age");  
17         }  
18     }  
19     catch (Exception e) {  
20         System.out.println("Registration failed");  
21     }  
22     finally {  
23         System.out.println("Registration completed");  
24     }  
25 }  
26 }  
27 }
```



Test Input

John
25



Output

Patient Name:JOHN
Adult patient
Registration completed

 Pending manual review

 Submitted